VALVE COMPONENTS

42 component records | Numeric values are numbers without units in the cell. Units: diameter mm; pressure bar; flow m^3/h. All numeric specifications are sample values for software testing, not vessel-approved operating data.

FA546 — STOP VALVE (GLOBE)

id	FA546
component_name	FA546 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA546
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	50
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA546_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA546_IN, PIPE_FA546_OUT
relationships.related_component_ids	

FA582 — BALL VALVE

id	FA582
component_name	FA582 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA582
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA582_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA582_IN, PIPE_FA582_OUT
relationships.related_component_ids	

FA572 — BALL VALVE

id	FA572
component_name	FA572 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA572
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	80
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA572_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA572_IN, PIPE_FA572_OUT
relationships.related_component_ids	

FA512 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA512
component_name	FA512 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA512
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA512_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA512_IN, PIPE_FA512_OUT
relationships.related_component_ids	

FA542 — GATE VALVE

id	FA542
component_name	FA542 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA542
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	52.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA542_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA542_IN, PIPE_FA542_OUT
relationships.related_component_ids	

FA541 — GATE VALVE

id	FA541
component_name	FA541 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA541
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	125
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	20.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA541_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA541_IN, PIPE_FA541_OUT
relationships.related_component_ids	

FA511 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA511
component_name	FA511 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA511
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	150
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA511_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA511_IN, PIPE_FA511_OUT
relationships.related_component_ids	

FA581 — BALL VALVE

id	FA581
component_name	FA581 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA581
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	50
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA581_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA581_IN, PIPE_FA581_OUT
relationships.related_component_ids	

FA571 — BALL VALVE

id	FA571
component_name	FA571 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA571
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA571_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA571_IN, PIPE_FA571_OUT
relationships.related_component_ids	

FA545 — STOP VALVE (GLOBE)

id	FA545
component_name	FA545 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA545
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	80
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA545_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA545_IN, PIPE_FA545_OUT
relationships.related_component_ids	

FA515 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA515
component_name	FA515 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA515
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	46.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA515_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA515_IN, PIPE_FA515_OUT
relationships.related_component_ids	

FA544 — GATE VALVE

id	FA544
component_name	FA544 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA544
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA544_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA544_IN, PIPE_FA544_OUT
relationships.related_component_ids	

FA514 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA514
component_name	FA514 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA514
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	125
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA514_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA514_IN, PIPE_FA514_OUT
relationships.related_component_ids	

FA574 — BALL VALVE

id	FA574
component_name	FA574 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA574
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	150
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA574_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA574_IN, PIPE_FA574_OUT
relationships.related_component_ids	

FA584 — BALL VALVE

id	FA584
component_name	FA584 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA584
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	50
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	36.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA584_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA584_IN, PIPE_FA584_OUT
relationships.related_component_ids	

FA548 — STOP VALVE (GLOBE)

id	FA548
component_name	FA548 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA548
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	65
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA548_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA548_IN, PIPE_FA548_OUT
relationships.related_component_ids	

FA543 — GATE VALVE

id	FA543
component_name	FA543 - GATE VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA543
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	80
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA543_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA543_IN, PIPE_FA543_OUT
relationships.related_component_ids	

FA513 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FA513
component_name	FA513 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA513
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	53.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA513_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA513_IN, PIPE_FA513_OUT
relationships.related_component_ids	

FA583 — BALL VALVE

id	FA583
component_name	FA583 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA583
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	100
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA583_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA583_IN, PIPE_FA583_OUT
relationships.related_component_ids	

FA573 — BALL VALVE

id	FA573
component_name	FA573 - BALL VALVE
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA573
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	125
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	36.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA573_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA573_IN, PIPE_FA573_OUT
relationships.related_component_ids	

FA547 — STOP VALVE (GLOBE)

id	FA547
component_name	FA547 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA547
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	150
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA547_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA547_IN, PIPE_FA547_OUT
relationships.related_component_ids	

FB544 — STOP VALVE (GLOBE)

id	FB544
component_name	FB544 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB544
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	50
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	13.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB544_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB544_IN, PIPE_FB544_OUT
relationships.related_component_ids	

FB582 — BALL VALVE

id	FB582
component_name	FB582 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB582
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	24.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB582_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB582_IN, PIPE_FB582_OUT
relationships.related_component_ids	

FB572 — BALL VALVE

id	FB572
component_name	FB572 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB572
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	80
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB572_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB572_IN, PIPE_FB572_OUT
relationships.related_component_ids	

FB512 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FB512
component_name	FB512 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB512
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB512_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB512_IN, PIPE_FB512_OUT
relationships.related_component_ids	

FB542 — GATE VALVE

id	FB542
component_name	FB542 - GATE VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB542
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	100
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	20.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB542_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB542_IN, PIPE_FB542_OUT
relationships.related_component_ids	

FB541 — GATE VALVE

id	FB541
component_name	FB541 - GATE VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB541
specifications.valve_type	GATE VALVE
specifications.nominal_diameter	125
specifications.material	Body: cast steel; gate: stainless steel; seat: metal
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	28.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Gate Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB541_F001
troubleshooting.failure_condition	Wedge stuck / seat passing
troubleshooting.symptoms	Valve passing when closed, handwheel hard to turn, stem leakage, high pressure drop when open.
troubleshooting.possible_causes	Wedge or seat wear, deposits, stem/nut wear, packing deterioration, corrosion.
troubleshooting.diagnostic_steps	Verify fully open/closed indication, inspect stem and gland, check travel without excessive force, compare line readings.
troubleshooting.immediate_action	Do not force handwheel or use as a throttling valve, inform engineer and isolate/depressurize before repair.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain stem and packing, avoid prolonged throttling, inspect and exercise periodically when permitted.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB541_IN, PIPE_FB541_OUT
relationships.related_component_ids	

FB511 — BUTTERFLY VALVE (LUG, GEARBOX TYPE)

id	FB511
component_name	FB511 - BUTTERFLY VALVE (LUG, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB511
specifications.valve_type	BUTTERFLY VALVE (LUG, GEARBOX TYPE)
specifications.nominal_diameter	150
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Lug type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Lug, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB511_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB511_IN, PIPE_FB511_OUT
relationships.related_component_ids	

FB571 — BALL VALVE

id	FB571
component_name	FB571 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB571
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	50
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	30.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB571_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB571_IN, PIPE_FB571_OUT
relationships.related_component_ids	

FB581 — BALL VALVE

id	FB581
component_name	FB581 - BALL VALVE
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB581
specifications.valve_type	BALL VALVE
specifications.nominal_diameter	65
specifications.material	Body: carbon steel; ball: stainless steel; seat: PTFE
specifications.connection_type	Flanged / screwed end
specifications.operation_type	Manual lever
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	36.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Ball Valve for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB581_F001
troubleshooting.failure_condition	Seat passing / stem seal leakage
troubleshooting.symptoms	Leakage through closed valve, stem/body leakage, stiff lever, incomplete quarter-turn, restricted flow.
troubleshooting.possible_causes	Damaged PTFE seat, debris in ball cavity, worn stem seal, corrosion, incorrect end-stop.
troubleshooting.diagnostic_steps	Confirm lever/position indicator, inspect stem and body, check for external leakage, compare line pressure and flow with normal values.
troubleshooting.immediate_action	Notify engineer, do not dismantle under pressure, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep clean, inspect seat and stem seals, operate through full travel when permitted, follow maker maintenance schedule.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB581_IN, PIPE_FB581_OUT
relationships.related_component_ids	

FB543 — STOP VALVE (GLOBE)

id	FB543
component_name	FB543 - STOP VALVE (GLOBE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB543
specifications.valve_type	STOP VALVE (GLOBE)
specifications.nominal_diameter	80
specifications.material	Body: cast steel; disc/stem: stainless steel; packing: graphite
specifications.connection_type	Flanged end
specifications.operation_type	Manual handwheel
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	8.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Stop Valve (Globe) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB543_F001
troubleshooting.failure_condition	Disc-seat leakage / stem packing leakage
troubleshooting.symptoms	Seat leakage, gland leakage, stiff handwheel, vibration/noise, unexpected pressure drop.
troubleshooting.possible_causes	Disc/seat wear, packing damage, deposits, bent stem, incorrect operation.
troubleshooting.diagnostic_steps	Check valve position, inspect bonnet and gland, verify smooth travel, compare upstream/downstream readings.
troubleshooting.immediate_action	Notify engineer, do not tighten or dismantle under pressure, follow approved isolation procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect packing and seating surfaces, operate smoothly, maintain per maker recommendations.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB543_IN, PIPE_FB543_OUT
relationships.related_component_ids	

FA516 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA516
component_name	FA516 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA516
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	25.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA516_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA516_IN, PIPE_FA516_OUT
relationships.related_component_ids	

FA521 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA521
component_name	FA521 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA521
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	32.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA521_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA521_IN, PIPE_FA521_OUT
relationships.related_component_ids	

FA520 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA520
component_name	FA520 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA520
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	125
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA520_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA520_IN, PIPE_FA520_OUT
relationships.related_component_ids	

FA518 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA518
component_name	FA518 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA518
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	150
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	46.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA518_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA518_IN, PIPE_FA518_OUT
relationships.related_component_ids	

FA519 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA519
component_name	FA519 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA519
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	50
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	53.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA519_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA519_IN, PIPE_FA519_OUT
relationships.related_component_ids	

FA517 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FA517
component_name	FA517 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Fuel Oil (F.O.) Filling & Transfer System
identification.location	Fuel Transfer Piping
identification.line_number	FA517
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	65
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	18.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Fuel Oil (F.O.) Filling & Transfer System.
troubleshooting.fault_code	FA575_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FA575_IN, PIPE_FA575_OUT
relationships.related_component_ids	

FA575 — SELF-CLOSING VALVE

id	FA575
component_name	FA575- SELF-CLOSING VALVE
component_type	Valve
identification.system_name	Marine Diesel Oil transfer branch (mock assignment; verify service)
identification.location	Fuel Transfer Piping
identification.line_number	FA575
specifications.valve_type	SELF-CLOSING VALVE
specifications.nominal_diameter	65
specifications.material	Body: bronze; stem: stainless steel; seat: PTFE
specifications.connection_type	Flanged end
specifications.operation_type	Spring-return self-closing mechanism
specifications.pressure_rating	10
specifications.valve_opening	100
specifications.flow_rate	15.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Self-Closing Valve for Marine Diesel Oil transfer branch (mock assignment; verify service).
troubleshooting.fault_code	65A_F001
troubleshooting.failure_condition	Return spring failure / incomplete closure
troubleshooting.symptoms	Fails to close, slow return, seat leakage, mechanism sticks open.
troubleshooting.possible_causes	Weak/broken spring, fouled hinge or stem, damaged seat, corrosion, incorrect installation.
troubleshooting.diagnostic_steps	Verify tag and service, inspect return mechanism, test only by approved method, inspect for leakage and obstruction.
troubleshooting.immediate_action	Notify engineer, never wedge or bypass the self-closing action, isolate and make safe before maintenance.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Keep mechanism clean, inspect spring and seat, follow maker instructions and planned maintenance.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_65A_IN, PIPE_65A_OUT
relationships.related_component_ids	

FB112 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FB112
component_name	FB112 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB112
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	32.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB112_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB112_IN, PIPE_FB112_OUT
relationships.related_component_ids	

FB111 — BUTTERFLY VALVE (WAFER, GEARBOX TYPE)

id	FB111
component_name	FB111 - BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB111
specifications.valve_type	BUTTERFLY VALVE (WAFER, GEARBOX TYPE)
specifications.nominal_diameter	100
specifications.material	Body: ductile iron; disc: stainless steel; seat: EPDM
specifications.connection_type	Wafer type
specifications.operation_type	Manual gearbox
specifications.pressure_rating	16
specifications.valve_opening	100
specifications.flow_rate	39.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Butterfly Valve (Wafer, Gearbox Type) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB111_F001
troubleshooting.failure_condition	Gearbox jam / disc-seat leakage
troubleshooting.symptoms	Disc/seat passing, external shaft leakage, stiff gearbox, incomplete opening or closing.
troubleshooting.possible_causes	Worn or contaminated seat, damaged disc edge, gearbox wear, shaft misalignment, loose flange bolts.
troubleshooting.diagnostic_steps	Check tag and line state, inspect body and shaft for leakage, compare gearbox indicator with disc position, verify upstream/downstream pressure using approved instruments.
troubleshooting.immediate_action	Stop operation if abnormal leakage or binding occurs, notify the responsible engineer, isolate and depressurize under vessel procedure before inspection. Do not force the gearbox.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Inspect gearbox, shaft, disc and seat, exercise only when permitted, follow maker lubrication and service intervals.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB111_IN, PIPE_FB111_OUT
relationships.related_component_ids	

FA001 — SAFETY / RELIEF VALVE (GLOBE / ANGLE)

id	FA001
component_name	FA001 - SAFETY / RELIEF VALVE (GLOBE / ANGLE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FB105
specifications.valve_type	SAFETY / RELIEF VALVE (GLOBE / ANGLE)
specifications.nominal_diameter	65
specifications.material	Body: cast steel; spring: alloy steel; trim: stainless steel
specifications.connection_type	Flanged inlet and discharge
specifications.operation_type	Spring-loaded automatic
specifications.pressure_rating	8
specifications.valve_opening	100
specifications.flow_rate	15.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Safety / Relief Valve (Globe / Angle) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB105_F001
troubleshooting.failure_condition	Failure to lift / failure to reseat
troubleshooting.symptoms	Unwanted discharge, failure to lift at set pressure, failure to reseat, chatter or outlet leakage.
troubleshooting.possible_causes	Incorrect set pressure, contaminated seat, spring deterioration, blocked discharge path, damaged trim.
troubleshooting.diagnostic_steps	Treat discharge as a safety event, verify pressure indication and discharge route, compare against certified set-pressure record. Do not adjust in service.
troubleshooting.immediate_action	Alert responsible engineer immediately, keep clear of discharge, never gag, isolate or adjust except under authorized procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain certification and inspection records, keep discharge path clear, arrange authorized periodic testing.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB105_IN, PIPE_FB105_OUT
relationships.related_component_ids	

FA002— SAFETY / RELIEF VALVE (GLOBE / ANGLE)

id	FA002
component_name	FA002 - SAFETY / RELIEF VALVE (GLOBE / ANGLE)
component_type	Valve
identification.system_name	Diesel Oil (D.O.) Filling & Transfer System
identification.location	Engine Room
identification.line_number	FA002
specifications.valve_type	SAFETY / RELIEF VALVE (GLOBE / ANGLE)
specifications.nominal_diameter	65
specifications.material	Body: cast steel; spring: alloy steel; trim: stainless steel
specifications.connection_type	Flanged inlet and discharge
specifications.operation_type	Spring-loaded automatic
specifications.pressure_rating	8
specifications.valve_opening	100
specifications.flow_rate	20.0
specifications.flow_pressure	4.0
specifications.pressure_inlet	4.2
specifications.pressure_outlet	4.0
usecase	Safety / Relief Valve (Globe / Angle) for Diesel Oil (D.O.) Filling & Transfer System.
troubleshooting.fault_code	FB104_F001
troubleshooting.failure_condition	Failure to lift / failure to reseat
troubleshooting.symptoms	Unwanted discharge, failure to lift at set pressure, failure to reseat, chatter or outlet leakage.
troubleshooting.possible_causes	Incorrect set pressure, contaminated seat, spring deterioration, blocked discharge path, damaged trim.
troubleshooting.diagnostic_steps	Treat discharge as a safety event, verify pressure indication and discharge route, compare against certified set-pressure record. Do not adjust in service.
troubleshooting.immediate_action	Alert responsible engineer immediately, keep clear of discharge, never gag, isolate or adjust except under authorized procedure.
troubleshooting.operational_impact	Reduced flow, Pressure loss
troubleshooting.verification_steps	Confirm no leakage, verify correct valve position, check line pressure/flow stability, obtain engineer acceptance after maintenance.
troubleshooting.prevention_measures	Maintain certification and inspection records, keep discharge path clear, arrange authorized periodic testing.
troubleshooting.related_component_ids	
simulation.current_status	CLOSED
simulation.current_position	0
simulation.commanded_position	0
simulation.flow_status	NO_FLOW
simulation.flow_rate	0
simulation.pressure_inlet	4.2
simulation.pressure_outlet	4.0
simulation.fault_status	false
simulation.fault_code	null
simulation.animation_state	IDLE
simulation.flow_state	false
simulation.operating_hours	0
relationships.connected_pipe_ids	PIPE_FB104_IN, PIPE_FB104_OUT
relationships.related_component_ids	